

Hartree-Fock theory: from the variational principle to $\langle a|\hat{f}|i\rangle = 0$, Thouless' theorem, and the stability of the mean field

FYS4480/FYS9480 – Week 39, September 24–25, 2026

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Thursday September 24: why a mean field; variational calculus and Lagrange multipliers; determinants under a change of basis; the Hartree-Fock equations by varying the coefficients; the density matrix and the self-consistent field; Brillouin's theorem; Hartree-Fock in second quantization – varying one orbital, $|\Phi_0\rangle + \eta a_{\mathbf{a}}^\dagger a_i |\Phi_0\rangle$, and the equation $\langle a|\hat{f}|i\rangle = 0$
Friday September 25: Thouless' theorem – what a nearby determinant is, and its proof; the stability of the Hartree-Fock solution – the energy to second order, the stability matrix, stationary versus minimal, and the Lipkin model as an explicit instability

Reading: these slides; lecture notes chapter 6, sections 6.1–6.7 and 6.10–6.11; Szabo and Ostlund chapter 3; Shavitt and Bartlett chapter 4. Code: `WeeklySlides/week39.ipynb`, `BookPrograms/chapter06/hartreefock.py`

Plan for week 39

Thursday September 24, 2×45 min

- Why a mean field; the two ingredients of Hartree-Fock theory; $E^{\text{HF}} \geq E_0$ (section 6.1)
- Variational calculus with Lagrange multipliers; determinants under a linear change of basis and the invariance under unitary mixing of the occupied orbitals (sections 6.2–6.3)
- The Hartree-Fock equations by varying the expansion coefficients; the Fock matrix; a nonlinear eigenvalue problem (section 6.4)
- The density matrix and the self-consistent field; convergence of the energy versus convergence of the orbitals; Brillouin's theorem (section 6.5)
- Hartree-Fock in second quantization: varying one orbital is adding a $1p\text{-}1h$ determinant; the energy to first order with complex η ; the normal-ordered Hamiltonian gives $\langle a | \hat{f} | i \rangle = 0$, and how that becomes the eigenvalue problem (section 6.6)

Friday September 25, 2×45 min

- Thouless' theorem: every determinant not orthogonal to $|c\rangle$ is $e^{\hat{T}}|c\rangle$ with $\hat{T}$ a sum of $1p\text{-}1h$ operators; the three steps of the proof; what it buys us (section 6.7)
- Stability: stationary is not minimal; the energy to second order; the stability matrix $\hat{M}$ with blocks Δ , A , B ; the criterion; $\hat{M}$ is the RPA matrix (section 6.10); the Lipkin model as an explicit instability, the pairing model as stable but useless (section 6.11)
- Exercises: suggested set for week 39, and answers to the week-38 set (last frames)

Thursday, lecture 1: Mean field, variations and the Hartree-Fock equations

Where we are, and the map of the week

Chapter 5 said

The exact ground state is $\sum_K C_K |\Phi_K\rangle$ over exponentially many determinants, and we cannot afford it. In a Hartree-Fock basis the reference decouples from the singles, and every truncation improves. **So: which single determinant is best?**

Chapter 6 answers, three times

1. Vary the *expansion coefficients* of the orbitals with Lagrange multipliers $\rightarrow$ the eigenvalue problem $\mathbf{h}^{\text{HF}} \mathbf{C} = \epsilon \mathbf{C}$ (6.4, Thursday)
2. Vary *one orbital* towards one unoccupied orbital $\rightarrow \langle a | \hat{f} | i \rangle = 0$ (6.6, Thursday)
3. Vary the *determinant* itself, with Thouless' theorem telling us what a nearby determinant is $\rightarrow$ the same, and the second-order term that decides stability (6.7, 6.10, Friday)

Many-body connection

Hartree-Fock is the foundation of everything after it: perturbation theory expands about it, coupled cluster exponentiates corrections to it, truncated CI is measured against it, and TDA/RPA turn out to be its second derivative. The condition $f_{ai} = 0$ derived today is why the singles are absent from the leading orders of all of them.

Notation as in weeks 35–38: $i, j, \dots \leq F$ holes, $a, b, \dots > F$ particles, $p, q, \dots$ any; $\langle pq | \hat{v} | rs \rangle$ antisymmetrised.

Why a mean field? (section 6.1)

Instead of adding determinants we keep exactly one and ask which one is best.

Hartree-Fock theory in two ingredients

A single-particle basis defined by its own eigenvalue problem,

$$\hat{h}^{\text{HF}} \psi_\alpha = \varepsilon_\alpha \psi_\alpha, \quad \hat{h}^{\text{HF}} = \hat{t} + \hat{u}_{\text{ext}} + \hat{u}^{\text{HF}},$$

and the prescription for the still unknown one-body potential $\hat{u}^{\text{HF}}$: choose it such that $E^{\text{HF}} = \langle \Phi_0 | \hat{H} | \Phi_0 \rangle$ is stationary, with $|\Phi_0\rangle$ the determinant of the N lowest solutions.

- The variational principle guarantees $E^{\text{HF}} \geq E_0$: an upper bound, like every Ritz value of week 38 – here with the *orbitals* as parameters and the expansion frozen at one term.
- We shall find that $\hat{h}^{\text{HF}}$ is exactly the operator $\hat{f}$ of the normal-ordered Hamiltonian of week 37,

$$\langle p | \hat{h}^{\text{HF}} | q \rangle = \langle p | \hat{f} | q \rangle = \langle p | \hat{t} + \hat{u}_{\text{ext}} | q \rangle + \sum_{i \leq F} \langle p i | \hat{v} | q i \rangle, \quad \langle p | \hat{u}^{\text{HF}} | q \rangle = \sum_{i \leq F} \langle p i | \hat{v} | q i \rangle.$$

- This is the whole content of the mean-field idea: **the interaction summed over the other fermions is an effective one-body field** in which a given fermion moves. It depends on which states are occupied (*medium dependence*) and on the interaction: different interactions give different mean fields at the same density.
- For atoms and molecules $\hat{u}_{\text{ext}}$ is the attraction of the nuclei. A nucleus has none: it is self-bound, and the entire one-body field must be generated by the sum over the occupied states.

Variational calculus and Lagrange multipliers (section 6.2)

The machinery of week 38, in its simplest setting: one particle, $E = \int \psi^* \hat{H} \psi d^3r$ with $\int \psi^* \psi d^3r = 1$ and $\hat{H} = -\frac{1}{2}\nabla^2 + V$. Integrating the kinetic term by parts, the constrained variation is

$$\delta \left\{ \int \left(\frac{1}{2} \nabla \psi^* \cdot \nabla \psi + V \psi^* \psi - \lambda \psi^* \psi \right) d^3r \right\} = 0.$$

With $f = \frac{1}{2}(\psi_x^* \psi_x + \psi_y^* \psi_y + \psi_z^* \psi_z) + V \psi^* \psi - \lambda \psi^* \psi$, the Euler-Lagrange equation for ψ^* ,

$$\frac{\partial f}{\partial \psi^*} - \frac{\partial}{\partial x} \frac{\partial f}{\partial \psi_x^*} - \frac{\partial}{\partial y} \frac{\partial f}{\partial \psi_y^*} - \frac{\partial}{\partial z} \frac{\partial f}{\partial \psi_z^*} = 0 \quad \implies \quad -\frac{1}{2}\nabla^2 \psi + V\psi = \lambda\psi.$$

- The multiplier is the energy and the stationarity condition is the Schrödinger equation. **The pattern:** a constrained variation produces an eigenvalue problem, and the multiplier enforcing normalisation is the eigenvalue. In FCI the unknowns were the coefficients C_K and the constraint $\sum |C_K|^2 = 1$; here the unknown is a function and the constraint an integral.
- In Hartree-Fock the unknowns are N orbitals and the constraints are the orthonormality integrals $\langle i|j \rangle = \delta_{ij}$. Naively that is $N(N+1)/2$ multipliers; the next slide explains why one per orbital is enough.

Pause to think: multipliers

Pause to think #1

(about 3 min – discuss with your neighbour)

1. In the one-particle example, why does the multiplier turn out to be the energy, and not some other number?
2. For N orbitals there are $N(N+1)/2$ independent orthonormality constraints $\langle i|j \rangle = \delta_{ij}$, but the Hartree-Fock equations of section 6.4 carry only N multipliers ϵ_i . What has been thrown away, and why is that allowed?

Answer to pause #1

1. Multiply the Euler-Lagrange equation $-\frac{1}{2}\nabla^2\psi + V\psi = \lambda\psi$ by ψ^* and integrate: $\lambda \int \psi^*\psi = \int \psi^*\hat{H}\psi = E$. With the constraint $\int \psi^*\psi = 1$ the multiplier is the expectation value of $\hat{H}$ in the stationary state – and since the stationary states are the eigenstates, that expectation value is the eigenvalue. The same argument gives $\lambda = E$ for the FCI coefficients in week 38.
2. The off-diagonal constraints $\langle i|j\rangle = 0, i \neq j$, have been thrown away. The full set of multipliers is a Hermitian matrix ϵ_{ij} and the equations read $\hat{h}^{\text{HF}}\psi_i = \sum_j \epsilon_{ij}\psi_j$. But the energy, and the determinant up to a phase, are invariant under a unitary mixing of the occupied orbitals (next slide), so we may rotate the occupied orbitals until ϵ_{ij} is diagonal. That rotation costs nothing and leaves one multiplier per orbital: the *canonical* Hartree-Fock orbitals. The diagonal form is a convention, not a consequence of the variation – a point that returns in section 6.6.

Determinants under a linear change of basis (section 6.3)

A determinant is linear in each column (and in each row) separately,

$$\begin{vmatrix} \alpha_1 b_{11} + \alpha_2 b_{12} & a_{12} \\ \alpha_1 b_{21} + \alpha_2 b_{22} & a_{22} \end{vmatrix} = \alpha_1 \begin{vmatrix} b_{11} & a_{12} \\ b_{21} & a_{22} \end{vmatrix} + \alpha_2 \begin{vmatrix} b_{12} & a_{12} \\ b_{22} & a_{22} \end{vmatrix}, \quad \text{and in general } \det(BC) = \det B \det C.$$

Expand new orbitals in a fixed orthonormal basis, $\psi_p(x) = \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x)$. The Slater determinant of the ψ 's is

$$\frac{1}{\sqrt{N!}} \begin{vmatrix} \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x_1) & \cdots & \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x_N) \\ \vdots & & \vdots \\ \sum_{\lambda} C_{t\lambda} \phi_{\lambda}(x_1) & \cdots & \sum_{\lambda} C_{t\lambda} \phi_{\lambda}(x_N) \end{vmatrix} = \det(C) \Phi(\phi),$$

with $\Phi(\phi)$ the determinant of the original basis functions (rows = orbitals, columns = particles, as in chapter 2). Two consequences, used constantly:

- **Orthonormality is preserved by a unitary C:** $w_j^{\dagger} w_i = v_j^{\dagger} U^{\dagger} U v_i = \delta_{ij}$. A unitary change of coefficients never spoils the basis.
- **A determinant is invariant, up to a phase, under any unitary mixing of its occupied orbitals:** it knows only which N -dimensional subspace they span. The Hartree-Fock solution is a statement about a **subspace**; the individual orbitals are fixed only by the convention that the Fock matrix be diagonal. Today's second derivation will use the same fact from the other side: a variation *inside* the occupied subspace does nothing.

The energy functional in terms of the coefficients (section 6.4)

The energy of a single determinant is the functional of week 34, E_{ref} of week 37:

$$E[\Phi] = \sum_{\mu=1}^N \langle \mu | \hat{h}_0 | \mu \rangle + \frac{1}{2} \sum_{\mu\nu=1}^N \langle \mu\nu | \hat{v} | \mu\nu \rangle_{\text{AS}}.$$

Insert $\psi_i = \sum_{\alpha} C_{i\alpha} \phi_{\alpha}$ (Greek indices over the whole fixed basis, Latin over the N occupied orbitals):

E as a function of the coefficients

$$E[\Phi^{\text{HF}}] = \sum_{i=1}^N \sum_{\alpha\beta} C_{i\alpha}^* C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle + \frac{1}{2} \sum_{i,j=1}^N \sum_{\alpha\beta\gamma\delta} C_{i\alpha}^* C_{j\beta}^* C_{i\gamma} C_{j\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{\text{AS}}.$$

- The coefficients are not free: $\langle i | j \rangle = \delta_{ij} = \sum_{\alpha} C_{i\alpha}^* C_{j\alpha}$. Introduce one multiplier ϵ_i per occupied orbital (pause #1) and make

$$F[\Phi^{\text{HF}}] = E[\Phi^{\text{HF}}] - \sum_{i=1}^N \epsilon_i \sum_{\alpha} C_{i\alpha}^* C_{i\alpha}$$

stationary. C and C^* may be varied independently (real and imaginary parts are two independent real variations): $\partial F / \partial C_{i\alpha}^* = 0$.

- Compare with FCI: there E was *quadratic* in the unknowns and the condition linear. Here E is *quartic*, and the condition will be an eigenvalue problem whose matrix depends on its own solution.

Carrying out the variation

Differentiate with respect to $C_{i\alpha}^*$. The one-body term gives $\sum_{\beta} C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle$. In the two-body term $C_{i\alpha}^*$ appears twice – as $C_{i\alpha}^*$ and as $C_{j\beta}^*$ with $j = i$ – with equal contributions by the symmetry of $\langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{AS}$ under the simultaneous interchange of both pairs; they cancel the $\frac{1}{2}$:

$$\sum_{\beta} C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle + \sum_{j=1}^N \sum_{\beta\gamma\delta} C_{j\beta}^* C_{j\delta} C_{i\gamma} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{AS} = \epsilon_i^{\text{HF}} C_{i\alpha}.$$

Relabel the dummy indices so that $C_{i\beta}$ factors out of both terms,
 $\sum_{\beta} \{ \langle \alpha | \hat{h}_0 | \beta \rangle + \sum_{j\gamma\delta} C_{j\gamma}^* C_{j\delta} \langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{AS} \} C_{i\beta} = \epsilon_i^{\text{HF}} C_{i\alpha}$:

The Hartree-Fock equations

$$\boxed{\sum_{\beta} h_{\alpha\beta}^{\text{HF}} C_{i\beta} = \epsilon_i^{\text{HF}} C_{i\alpha}} \quad h_{\alpha\beta}^{\text{HF}} = \langle \alpha | \hat{h}_0 | \beta \rangle + \sum_{j=1}^N \sum_{\gamma\delta} C_{j\gamma}^* C_{j\delta} \langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{AS}.$$

- A Hermitian eigenvalue problem, $h^{\text{HF}} C = \epsilon^{\text{HF}} C$, with the one complication that **the matrix depends on its own eigenvectors**. In the eigenbasis of $\hat{h}_0$ the first term is $\epsilon_{\alpha} \delta_{\alpha\beta}$. No integrals are recomputed: $\langle \alpha | \hat{h}_0 | \beta \rangle$ and $\langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{AS}$ are tabulated once (n^4 storage), each iteration is a contraction with the current coefficients and a diagonalisation.

What the single-particle energies are, and are not

- In the Hartree-Fock basis the Fock matrix is diagonal with eigenvalues $\varepsilon_p^{\text{HF}} = \langle p | \hat{h}_0 | p \rangle + \sum_{j \leq F} \langle pj | \hat{v} | pj \rangle$. Summing over the occupied orbitals *double counts* the interaction:

$$\sum_{i \leq F} \varepsilon_i^{\text{HF}} = \sum_i \langle i | \hat{h}_0 | i \rangle + \sum_{ij} \langle ij | \hat{v} | ij \rangle \quad \Rightarrow \quad \boxed{E^{\text{HF}} = \sum_{i \leq F} \varepsilon_i^{\text{HF}} - \frac{1}{2} \sum_{ij \leq F} \langle ij | \hat{v} | ij \rangle}$$

(chapter 6, exercise 2b; 1.29801640 both ways for the random Hamiltonian of week 38).

- Koopmans' theorem** (exercise 2c): remove a particle from orbital k *without* letting the other orbitals relax. The energy of the $N - 1$ determinant is $E^{\text{HF}} - \varepsilon_k^{\text{HF}}$, so $-\varepsilon_k$ is the (frozen-orbital) removal energy – an ionisation energy for atoms, a separation energy for nuclei. Relaxation and correlations correct it; for closed-shell atoms it is nevertheless within a few per cent.
- The unoccupied $\varepsilon_a^{\text{HF}}$ are *not* affinities of the same quality: an added particle feels the field of N others, a removed one of $N - 1$. The asymmetry is in the sum over $j \leq F$, which is the same for both.

Many-body connection

E^{HF} and the $\varepsilon_p^{\text{HF}}$ are functionals of the occupied subspace only: a unitary mixing of the occupied orbitals changes neither (section 6.3 once more).

Thursday, lecture 2: The self-consistent field, and Hartree-Fock in second quantization

The density matrix and the self-consistent field (section 6.5)

The double sum over occupied orbitals enters only through the one-body **density matrix**

$$\rho_{\gamma\delta} = \sum_{i=1}^N \langle \gamma | i \rangle \langle i | \delta \rangle = \sum_{i=1}^N C_{i\gamma} C_{i\delta}^*, \quad \rho^2 = \rho, \quad \text{tr } \rho = N, \quad h_{\alpha\beta}^{\text{HF}} = \varepsilon_{\alpha} \delta_{\alpha\beta} + \sum_{\gamma\delta} \rho_{\gamma\delta} \langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{\text{AS}},$$

the projector onto the occupied subspace (section 6.3 formalised: h^{HF} depends on the subspace, not on the orbitals spanning it).

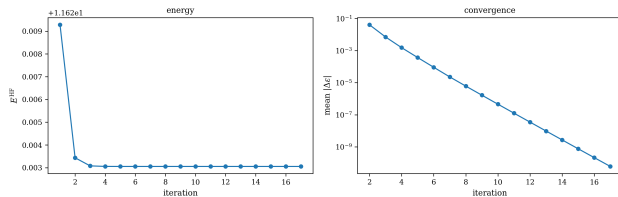
The self-consistent field algorithm

1. Start from a guess, conventionally $C_{i\alpha}^{(0)} = \delta_{i\alpha}$ (the N lowest eigenstates of $\hat{h}_0$); random orthonormal vectors also work.
2. Build ρ and h^{HF} ; diagonalise; take the N lowest eigenvectors as the new occupied orbitals.
3. Repeat until the single-particle energies stop moving, $\frac{1}{m} \sum_p |\epsilon_p^{(n)} - \epsilon_p^{(n-1)}| \leq \lambda \sim 10^{-8}$.

`hartreefock.py`, spinless fermions in a 1D oscillator trap with a softened Coulomb repulsion, eight orbitals; the gain grows with N :

N	iterations	$E(\text{reference})$	$E(\text{Hartree-Fock})$	gain
2	17	2.69018342	2.66308878	-2.71×10^{-2}
3	17	6.46801021	6.39016133	-7.78×10^{-2}
4	17	11.77084712	11.62305385	-1.48×10^{-1}

Watching it converge: the energy is not the right criterion



$N = 4$ in the trap. Left: E^{HF} after each diagonalisation (offset 11.62). Right: the mean drift of the single-particle energies, logarithmic.

- The first diagonalisation already gives 96% of the total gain; after three iterations the energy is flat on any resolvable scale.
- The orbitals are nowhere near converged: the drift falls by a fixed factor of about four per iteration, from 4×10^{-2} to 10^{-10} in seventeen steps – **linear convergence** of a fixed-point iteration, rate set by the spectral radius of the map.
- Why the energy looks better: at a stationary point it is *quadratic* in the orbital displacement. A drift of 10^{-5} in the ϵ_p shows up as 10^{-10} in E .

Never use the energy as the convergence criterion. Everything that depends on the orbitals – f_{ai} , transition matrix elements, the input of a coupled-cluster calculation – is only as good as the right-hand panel.

Brillouin's theorem, read off from a converged calculation

At convergence the Fock matrix is diagonal in its own eigenbasis, so its occupied-unoccupied block vanishes:

$$f_{ai} = 0 \quad \Longleftrightarrow \quad \langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0 \quad (\text{the program reports } \max |f_{ai}| = 8.7 \times 10^{-11} \text{ for } N = 4).$$

The Hartree-Fock solution is **the single-particle basis that decouples the reference from all single excitations**.
Week 38, random Hamiltonian, $n = 8$, $N = 4$:

basis	E_{ref} or E^{HF}	CIS	CID	CISD	FCI
eigenstates of $\hat{h}_0$	6.497174	2.438662	5.442075	1.157492	0.747713
Hartree-Fock (48 iterations)	1.298016	1.298016	0.819738	0.774941	0.747713

- CIS returns E^{HF} exactly, the $0p0h-1p1h$ block of the CI matrix is empty, and the correlation energy -0.550 is carried by the doubles term of the energy equation alone. Every truncation is better in the HF basis; the full CI is the same.
- So far Brillouin's theorem is an *observation* about the converged Fock matrix. The second half of this lecture derives it from the variational principle, and shows that it *is* the Hartree-Fock equation.

Pause to think: what converges, and why

Pause to think #2

(about 3 min – discuss with your neighbour)

1. The energy settles in three iterations, the orbitals need seventeen. Use the variational principle to explain why an error δ in the orbitals gives an error of order δ^2 in the energy, and give one quantity for which the error stays of order δ .
2. The loop “build ρ , diagonalise, repeat” is a fixed-point iteration $C^{(n+1)} = G(C^{(n)})$. Why does that produce a straight line on the logarithmic plot? What would you expect from Newton’s method instead?

Answer to pause #2

1. At a stationary point the first-order change of E under any admissible variation vanishes – that is what stationary means – so $E(\Phi_0 + \delta\Phi) = E^{\text{HF}} + \mathcal{O}(\delta^2)$. Any quantity that is *not* stationary at the solution, such as an off-diagonal Fock matrix element f_{ai} , a transition matrix element $\langle \Phi_0 | \hat{O} | \Phi_i^a \rangle$, or the amplitudes of a subsequent perturbation or coupled-cluster calculation, retains an error of order δ . This is why the drift of the ϵ_p , not the energy, is the convergence criterion.
2. Near the fixed point C^* the map is linear, $C^{(n+1)} - C^* \simeq G'(C^*)(C^{(n)} - C^*)$, so the error is multiplied by the spectral radius $\rho(G')$ at every step: $\log(\text{error})$ decreases linearly with n , here with slope $\log(1/4)$. Newton's method (or DIIS, the standard accelerator) uses the derivative of the map and converges quadratically: the number of correct digits doubles per step, and the same plot would curve down ever more steeply. Chapter 1, section on iterative solvers.

Hartree-Fock in second quantization: a shorter route (section 6.6)

Section 6.4 varied *all* the coefficients of *all* the occupied orbitals at once and needed Lagrange multipliers to keep them orthonormal. We now vary a **single** occupied orbital by a **single** amount, and ask what that does to the determinant.

- In first quantization the answer is an elementary property of determinants (linearity in one row).
- In second quantization it says that the varied determinant is a $1p\text{-}1h$ excitation of the original one.
- The stationarity condition then collapses to *one* matrix element of the Fock operator,

$$\langle a | \hat{f} | i \rangle = 0 \quad \text{for all } i \leq F, \ a > F,$$

which is the Hartree-Fock equation in its most compact form – the form Brillouin's theorem asserts, the form that empties the $0p0h\text{-}1p1h$ block of the CI matrix, and the form that removes the singles from the leading orders of perturbation theory and coupled cluster.

Notation for the rest of the lecture

$|\Phi_0\rangle$ a normalised determinant of N orbitals $\phi_1 \dots \phi_N$ from an orthonormal basis $\{\phi_\lambda\}$; $i, j \leq F$ occupied, $a, b > F$ unoccupied; $|\Phi_i^a\rangle = a_a^\dagger a_i |\Phi_0\rangle$. Nothing assumes that $|\Phi_0\rangle$ is the Hartree-Fock determinant: we are looking for the condition that makes it one.

Varying one orbital

$$\Phi_0 = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(x_1) & \cdots & \phi_1(x_N) \\ \vdots & \ddots & \vdots \\ \phi_N(x_1) & \cdots & \phi_N(x_N) \end{vmatrix}, \quad \phi_i(x) \longrightarrow \phi'_i(x) = N_i(\phi_i(x) + \eta \phi_a(x)), \quad |\eta| \ll 1, \quad \eta \in \mathbb{C}.$$

Why this is the general case

- An arbitrary $\delta\phi_i$ can be expanded in the basis. Its component along an occupied ϕ_j , $j \neq i$, adds – by linearity in row i – a determinant with *two identical rows*, which vanishes; its component along ϕ_i itself only rescales. **Variations inside the occupied subspace do nothing** (section 6.3 from the other side). Only components towards *unoccupied* orbitals change the state.
- A variation towards several unoccupied orbitals is a sum of variations towards one. So one pair (i, a) at a time suffices.
- Normalisation: $N_i^{-2} = \int dx (\phi_i^* + \eta^* \phi_a^*)(\phi_i + \eta \phi_a) = 1 + |\eta|^2$, the cross terms vanishing by orthogonality, so $N_i = 1 + \mathcal{O}(|\eta|^2)$ and we may set $N_i = 1$ to first order.
- The varied orbital is automatically orthogonal to the other occupied ones, because ϕ_i and ϕ_a both are. **This is where the Lagrange multipliers have gone**: by varying towards an orbital orthogonal to the occupied space, orthonormality is built into the variation instead of being enforced afterwards.

The varied determinant is a $1p\text{-}1h$ excitation

A determinant is linear in each row. Inserting $\phi_i + \eta\phi_a$ into row i ,

$$\Phi_0 \longrightarrow \Phi'_0 = N_i(\Phi_0 + \eta\Phi_i^a), \quad \Phi_i^a = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(x_1) & \cdots & \phi_1(x_N) \\ \vdots & & \vdots \\ \phi_a(x_1) & \cdots & \phi_a(x_N) \\ \vdots & & \vdots \\ \phi_N(x_1) & \cdots & \phi_N(x_N) \end{vmatrix} \quad (\phi_a \text{ in row } i),$$

a normalised determinant in its own right, orthogonal to Φ_0 (the two share only $N - 1$ orbitals).

In second quantization: one line, and a sign to check

$|\Phi_0\rangle = a_1^\dagger a_2^\dagger \cdots a_i^\dagger \cdots a_N^\dagger |0\rangle$. Replacing ϕ_i by ϕ_a means replacing $a_i^\dagger$ by $a_a^\dagger$ in the i -th slot – and that is what $a_a^\dagger a_i$ does: moving a_i to the right through the $i - 1$ operators in front of $a_i^\dagger$ costs $(-1)^{i-1}$; $a_i a_i^\dagger = 1 - a_i^\dagger a_i$ removes $a_i^\dagger$ (the second term kills the vacuum); moving $a_a^\dagger$ into the vacated slot costs $(-1)^{i-1}$ again. Net sign: none.

$$|\Phi_i^a\rangle = a_a^\dagger a_i |\Phi_0\rangle = a_1^\dagger \cdots a_a^\dagger \cdots a_N^\dagger |0\rangle,$$

$$|\Phi'_0\rangle = |\Phi_0\rangle + \eta a_a^\dagger a_i |\Phi_0\rangle + \mathcal{O}(|\eta|^2)$$

$$\langle \Phi_i^a | \Phi_0 \rangle = 0.$$

Varying an occupied orbital towards an unoccupied one is, to first order, adding a one-particle-one-hole excitation to the determinant. Everything that follows rests on this. (In the bit representation of week 38 the orbitals are stored in ascending order, so re-ordering $a_a^\dagger$ costs $(-1)^{\#\text{occupied between } i \text{ and } a}$ – the notebook checks it.)

The energy to first order

With N_i kept the energy of the varied determinant is exact,

$$E(\eta) = \frac{\langle \Phi'_0 | \hat{H} | \Phi'_0 \rangle}{\langle \Phi'_0 | \Phi'_0 \rangle} = \frac{E_0 + \eta^* \langle \Phi_i^a | \hat{H} | \Phi_0 \rangle + \eta \langle \Phi_0 | \hat{H} | \Phi_i^a \rangle + |\eta|^2 \langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle}{1 + |\eta|^2}, \quad E_0 = \langle \Phi_0 | \hat{H} | \Phi_0 \rangle,$$

and expanding the denominator,

$$E(\eta) = E_0 + \eta^* \langle \Phi_i^a | \hat{H} | \Phi_0 \rangle + \eta \langle \Phi_0 | \hat{H} | \Phi_i^a \rangle + |\eta|^2 (\langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle - E_0) + \mathcal{O}(|\eta|^3).$$

Stationarity

$|\Phi_0\rangle$ is a Hartree-Fock solution if $E(\eta)$ is stationary at $\eta = 0$ for every i, a and η : the terms linear in η must vanish. With $z = \langle \Phi_0 | \hat{H} | \Phi_i^a \rangle$ and $\hat{H}$ Hermitian they add up to $\eta z + \eta^* z^* = 2 \operatorname{Re}(\eta z)$. Choosing η real forces $\operatorname{Re} z = 0$; choosing it purely imaginary forces $\operatorname{Im} z = 0$. Hence

$$\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0 \quad \text{and} \quad \langle \Phi_i^a | \hat{H} | \Phi_0 \rangle = 0 \quad \text{for all } i \leq F, a > F.$$

- A real η alone would only have fixed the real part of the matrix element – this is why the variation had to be complex.
- The $|\eta|^2$ term is not needed for stationarity, but note what it is: along this one direction the second variation is the excitation energy of the $1p$ - $1h$ determinant relative to the reference. Friday does this for all directions at once.

Pause to think: real or complex

Pause to think #3

(about 3 min – discuss with your neighbour)

1. For a real Hamiltonian and real orbitals, $z = \langle \Phi_0 | \hat{H} | \Phi_i^a \rangle$ is real. Does a real η then suffice? What is lost in general if one insists on real variations, and what is the analogue in the coefficient derivation of section 6.4?
2. Sketch $E(\eta)$ for real η in two cases: $|\Phi_0\rangle$ an arbitrary determinant, and $|\Phi_0\rangle$ the Hartree-Fock determinant. What is the slope, what is the curvature?

Answer to pause #3

1. If everything is real, $\text{Im } z = 0$ automatically and a real η suffices. In general the condition $\text{Re}(\eta z) = 0$ for real η only gives $\text{Re } z = 0$, and the imaginary part of f_{ia} would be left unconstrained: the determinant would be stationary under real rotations only, which is not stationary. In section 6.4 the analogue is treating C and C^* as independent variables – equivalent to varying real and imaginary parts separately – and imposing $\partial F / \partial C_{i\alpha}^* = 0$, which is one complex equation, i.e. two real ones.
2. Arbitrary $|\Phi_0\rangle$: a straight line through E_0 with slope $2f_{ia}$, bent by the quadratic term – the energy can be lowered by moving along $-\text{sgn}(f_{ia})$. Hartree-Fock $|\Phi_0\rangle$: no linear term, a parabola with its minimum at $\eta = 0$ and curvature $\langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle - E_0$ (for a stable solution). The notebook draws both for the trap system: slope $2f_{ia} = -0.867$ at $(a, i) = (4, 2)$ in the oscillator basis, -1.7×10^{-10} in the Hartree-Fock basis.

The matrix element, with the normal-ordered Hamiltonian

Write $\hat{H}$ in the normal-ordered form of week 37, **taken with respect to $|\Phi_0\rangle$ itself** – any determinant defines a normal ordering and a Fock matrix of its own, with the sum over j running over *its* occupied orbitals:

$$\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N, \quad \hat{F}_N = \sum_{pq} f_{pq} \{a_p^\dagger a_q\}, \quad f_{pq} = \langle p | \hat{h}_0 | q \rangle + \sum_{j \leq F} \langle pj | \hat{v} | qj \rangle_{\text{AS}}.$$

- $E_{\text{ref}} \langle \Phi_0 | \Phi_i^a \rangle = 0$.
- $\hat{V}_N$ is a normal-ordered group of four operators; $a_a^\dagger a_i$ supplies two. The generalised Wick theorem would have to contract two operators of $\hat{V}_N$ with each other, which it forbids: **zero** (the argument that gave $\langle \Phi_i^a | \hat{H} | c \rangle = f_{ai}$ in week 37).
- $\hat{F}_N$ survives, and needs no Wick's theorem. Since $\{a_p^\dagger a_q\} = a_p^\dagger a_q - \langle \Phi_0 | a_p^\dagger a_q | \Phi_0 \rangle$ and the subtracted number multiplies $\langle \Phi_0 | a_a^\dagger a_i | \Phi_0 \rangle = 0$,

$$\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = \sum_{pq} f_{pq} \langle \Phi_0 | a_p^\dagger a_q a_a^\dagger a_i | \Phi_0 \rangle.$$

Read from the right: a_i empties i , $a_a^\dagger$ fills a ; to return to $|\Phi_0\rangle$ the remaining two must undo exactly this, so $q = a$, $p = i$, every other choice leaving a state orthogonal to the reference. Then $a_a a_a^\dagger = 1 - a_a^\dagger a_a$, the second term vanishes on $a_i | \Phi_0 \rangle$ (a is empty there), and $\langle \Phi_0 | a_i^\dagger a_i | \Phi_0 \rangle = 1$.

Result

$$\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = f_{ia} = \langle i | \hat{f} | a \rangle, \quad \langle \Phi_i^a | \hat{H} | \Phi_0 \rangle = f_{ai} = f_{ia}^* = \langle a | \hat{f} | i \rangle.$$

The Hartree-Fock equation: $\langle a|\hat{f}|i\rangle = 0$

The stationarity condition therefore reads

$$\langle a|\hat{f}|i\rangle = \langle a|\hat{h}_0|i\rangle + \sum_{j \leq F} \langle aj|\hat{v}|ij\rangle_{AS} = 0 \quad \text{for all } i \leq F, a > F$$

- This is the Hartree-Fock equation. The Fock operator built from a determinant must have no matrix elements between the orbitals it occupies and those it leaves empty. Comparing with section 6.1, $\hat{f} = \hat{h}^{\text{HF}} = \hat{h}_0 + \hat{u}^{\text{HF}}$, and the mean field $\hat{u}^{\text{HF}}$ is simply the interaction part of f_{pq} .

- Equivalently, in the language of determinants,

$$\langle \Phi_0^{\text{HF}} | \hat{H} | \Phi_i^a \rangle = 0, \quad | \Phi_i^a \rangle = a_a^\dagger a_i | \Phi_0^{\text{HF}} \rangle :$$

Brillouin's theorem, now derived from the variational principle rather than read off from a converged calculation. The reference does not couple to any singly excited determinant.

- Coupling to doubles is *not* forbidden: $\langle \Phi_0^{\text{HF}} | \hat{H} | \Phi_{ij}^{ab} \rangle = \langle ij|\hat{v}|ab\rangle$. This is the block structure of the CI matrix of week 38 – the reference row has a zero in the 1p-1h block and non-zero entries in the 2p-2h block – and all the correlation energy that Hartree-Fock misses enters through the latter.

	ref	1p1h	2p2h	higher
ref	E^{HF}	0	×	0
1p1h	0	×	×	×
2p2h	×	×	×	×

From the condition to the eigenvalue problem

$\langle a|\hat{f}|i\rangle = 0$ contains no eigenvalues; section 6.4 gave an eigenvalue problem. The step between them is the freedom of section 6.3. Order the basis with the occupied orbitals first: the condition says the Fock matrix is **block diagonal**,

$$f = \begin{pmatrix} f_{oo} & 0 \\ 0 & f_{uu} \end{pmatrix}, \quad (f_{oo})_{ij} = f_{ij}, \quad (f_{uu})_{ab} = f_{ab},$$

with the diagonal blocks unconstrained.

- Each block is Hermitian and can be diagonalised by a unitary matrix. Diagonalising f_{oo} mixes the occupied orbitals among themselves, which changes $|\Phi_0\rangle$ by a phase only; diagonalising f_{uu} does not touch the determinant at all; neither alters $f_{ai} = 0$. In the resulting basis

$$\begin{aligned} \langle p|q\rangle &= \delta_{pq}, & \langle i|\hat{f}|a\rangle &= \langle a|\hat{f}|i\rangle = 0, & \langle i|\hat{f}|i\rangle &= \varepsilon_i^{\text{HF}}, & \langle a|\hat{f}|a\rangle &= \varepsilon_a^{\text{HF}}, \\ \hat{f}|p\rangle &= \varepsilon_p^{\text{HF}}|p\rangle, & \langle p|\hat{f}|q\rangle &= \varepsilon_p^{\text{HF}}\delta_{pq}, & |\Phi_0^{\text{HF}}\rangle &= \prod_{i \leq F} a_i^\dagger |0\rangle, \end{aligned}$$

which is $h^{\text{HF}}C = \epsilon C$ written in the basis it defines. Conversely, if $\hat{f}$ is diagonal its occupied-unoccupied block vanishes trivially.

- So the eigenvalue form = the variational condition + a convention: **the Hartree-Fock state is fixed by $f_{ai} = 0$, the Hartree-Fock orbitals by the choice that $\hat{f}$ be diagonal.**
- This is what makes the SCF loop a correct algorithm: it diagonalises f at every step, and at its fixed point f is diagonal in the basis that generated it, so $f_{ai} = 0$ to the tolerance the loop was run to – the $\max |f_{ai}|$ reported by the program is a direct measure of how well the Hartree-Fock equation is satisfied.

The same calculation in the coefficient language

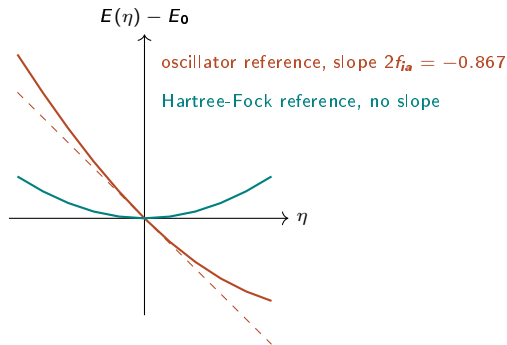
The variation $\phi_i \rightarrow \phi_i + \eta \phi_a$ is not a different variation from section 6.4, it is the same one organised differently: in the fixed basis it is the change $\delta C_{i\lambda} = \eta C_{a\lambda}$ of the coefficients of orbital i , along the coefficient vector of orbital a .

- Section 6.4 let $\delta C_{i\lambda}$ point anywhere and paid with multipliers, which became the eigenvalues. Here the direction is restricted to the orthogonal complement of the occupied space; no multiplier, no eigenvalue – until f is diagonalised by choice.
- Project $\sum_{\beta} h_{\alpha\beta}^{\text{HF}} C_{i\beta} = \epsilon_i C_{i\alpha}$ on the coefficient vector of orbital a : the right-hand side gives $\epsilon_i \sum_{\alpha} C_{a\alpha}^* C_{i\alpha} = 0$, the multiplier drops out precisely because the direction is orthogonal to the occupied space, and what remains is $\langle a | \hat{f} | i \rangle = 0$.

Numerical check

OrbitalVariation in hartreefock.py, four fermions in eight trap orbitals, all checked in the 70-determinant space:

- $a_a^\dagger a_i |\Phi_0\rangle$ is the bit pattern with $i \rightarrow a$ for all 16 pairs (sign as predicted)
- $\max |\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle - f_{ia}| = 2 \times 10^{-15}$
- oscillator reference: $\max |f_{ai}| = 0.43$; odd part of $E(\eta)$ at $\eta = 0.05$: -4.33×10^{-2} against $2\eta f_{ia} = -4.34 \times 10^{-2}$; imaginary η : no linear change (real f_{ia})
- Hartree-Fock reference: $\max |f_{ai}| = 8.7 \times 10^{-11}$, odd part below 10^{-11} – stationary; what is left is the quadratic term



$E(\eta) - E_0$ for real η and the pair $(a, i) = (4, 2)$: a line through E_0 bent by the quadratic term, and a parabola with its minimum at $\eta = 0$.

The general variation, and what is still missing

A general first-order variation of the determinant is a superposition of the single ones,

$$|\Phi'_0\rangle = |\Phi_0\rangle + \sum_{a>F} \sum_{i\leq F} \eta_{ai} a_a^\dagger a_i |\Phi_0\rangle + \mathcal{O}(\eta^2),$$

and since the $1p\text{-}1h$ determinants are mutually orthogonal and orthogonal to $|\Phi_0\rangle$, stationarity under it is stationarity under each term separately: $\langle a|\hat{f}|i\rangle = 0$ for every pair is the **complete** first-order condition.

Two questions, both needing second order

1. Are the $1p\text{-}1h$ excitations really all there is – does the family of determinants near $|\Phi_0\rangle$ have exactly $(n - N) \times N$ complex parameters, and no others?
2. Is the stationary point a minimum?

Answering them means writing down the *exact* form of an arbitrary nearby determinant, not merely its first-order approximation. That is Thouless' theorem – Friday.

Many-body connection

Three derivations of one equation: coefficients with multipliers give $h^{\text{HF}}C = \epsilon C$; one orbital at a time gives $f_{ai} = 0$; the determinant as a whole will give $f_{ai} = 0$ plus the stability matrix.

Friday, lecture 3: Thouless' theorem

Where Thursday ended, and what a nearby determinant is

- The ansatz: $|\Phi_0\rangle = |c\rangle = a_i^\dagger a_j^\dagger \cdots a_l^\dagger |0\rangle$, and we seek the determinant for which $E_0 = \langle c|\hat{H}|c\rangle$ is a local minimum.
- Stationarity: $\langle \delta c|\hat{H}|c\rangle = 0$ for every *admissible* $|\delta c\rangle$ – every variation that keeps $|c\rangle + |\delta c\rangle$ a Slater determinant. Thursday used $|\delta c\rangle = \eta a_a^\dagger a_i |c\rangle$ to first order. To go further we need to know what the neighbourhood of a determinant looks like *inside the set of determinants*.

Theorem (Thouless, 1960)

Any Slater determinant $|c'\rangle$ that is not orthogonal to a given determinant $|c\rangle = \prod_{i \leq F} a_i^\dagger |0\rangle$ can be written as

$$|c'\rangle = \exp\left\{\sum_{a > F} \sum_{i \leq F} C_{ai} a_a^\dagger a_i\right\} |c\rangle = e^{\hat{T}} |c\rangle.$$

In words: every determinant in the neighbourhood of $|c\rangle$ is obtained from it by an exponentiated linear combination of $1p$ - $1h$ excitations. The manifold of Slater determinants is parametrised, near any point, by the $(n - N) \times N$ complex amplitudes C_{ai} **and by nothing else** – so the Hartree-Fock variation has exactly that many degrees of freedom, and the $2p$ - $2h$, $3p$ - $3h$, ... admixtures that a general state has are all generated by the exponential. The proof has three steps: the exponential collapses; the result is a determinant; every determinant is of this form.

Step one: the exponential collapses

The exponential of a sum factorises into a product of exponentials only when the operators commute (otherwise Baker-Campbell-Hausdorff, section 6.8, supplies the correction). Here we are lucky. For $a, b > F$ and $i, j \leq F$ the index sets are disjoint, so $a_i a_b^\dagger = -a_b^\dagger a_i$ and $a_j a_a^\dagger = -a_a^\dagger a_j$, and both orderings reduce to $-a_a^\dagger a_b^\dagger a_i a_j$:

$$[a_a^\dagger a_i, a_b^\dagger a_j] = 0.$$

Write $\hat{T} = \sum_{i \leq F} \hat{T}_i$ with $\hat{T}_i = \sum_{a > F} C_{ai} a_a^\dagger a_i$. Then $e^{\hat{T}} = \prod_i e^{\hat{T}_i}$, and each factor truncates, because

$$\hat{T}_i^2 = \sum_{ab > F} C_{ai} C_{bi} a_a^\dagger a_i a_b^\dagger a_i = 0$$

– the rightmost a_i empties hole i and the second one annihilates the state ($a_i^2 = 0$). Hence

$$|c'\rangle = \prod_{i \leq F} \left\{ 1 + \sum_{a > F} C_{ai} a_a^\dagger a_i \right\} |c\rangle.$$

This is not $1 + \hat{T}$. The cross terms $\hat{T}_i \hat{T}_j$, $i \neq j$, survive and are $2p$ - $2h$ admixtures; $\hat{T}_i \hat{T}_j \hat{T}_k$ are $3p$ - $3h$, and so on up to Np - Nh . They are second and higher order in the amplitudes, which is why dropping them is legitimate for an *infinitesimal* variation – exactly what Thursday did, and what the stability analysis will do at the next order.

Step two: the product is a single determinant

Insert $|c\rangle = a_{i_1}^\dagger a_{i_2}^\dagger \cdots a_{i_N}^\dagger |0\rangle$. The factor labelled i_1 acts only on $a_{i_1}^\dagger$ (it commutes with every other creation operator and annihilates the vacuum), the factor labelled i_2 only on $a_{i_2}^\dagger$, and so on:

$$|c'\rangle = \left(1 + \sum_a C_{ai_1} a_a^\dagger a_{i_1}\right) a_{i_1}^\dagger \left(1 + \sum_a C_{ai_2} a_a^\dagger a_{i_2}\right) a_{i_2}^\dagger \cdots |0\rangle = \prod_{i \leq F} \left(a_i^\dagger + \sum_{a > F} C_{ai} a_a^\dagger\right) |0\rangle.$$

New creation operators

$$b_i^\dagger = a_i^\dagger + \sum_{a > F} C_{ai} a_a^\dagger, \quad |c'\rangle = \prod_{i \leq F} b_i^\dagger |0\rangle.$$

- $|c'\rangle$ has exactly the form of a Slater determinant: **the rotated state is a single determinant, not a superposition**. This is the whole point.
- The new orbitals $\psi_i = \phi_i + \sum_a C_{ai} \phi_a$ are Thursday's variation for all (i, a) at once – with the exponential supplying, automatically, the products that make the result an exact determinant instead of a first-order approximation.
- They are not normalised ($\langle \psi_i | \psi_j \rangle = \delta_{ij} + \sum_a C_{ai}^* C_{aj}$) and $\langle c | c' \rangle = 1$: this is *intermediate normalisation*, the same convention as for the FCI amplitudes of week 38. A unitary mixing of the $b_i^\dagger$ changes the determinant by a factor only (section 6.3), so nothing is lost.

Step three: it is the general determinant

$b_i^\dagger$ looks less general than it should, because the sum runs only over $a > F$. Take the most general new orbitals,

$$\tilde{b}_i^\dagger = \sum_p f_{ip} a_p^\dagger, \quad |\tilde{c}\rangle = \prod_{i \leq F} \tilde{b}_i^\dagger |0\rangle,$$

with f unitary, and require only $\langle c|\tilde{c}\rangle \neq 0$. The overlap is

$$\langle c|\tilde{c}\rangle = \langle 0|a_{i_N} \cdots a_{i_1} \left(\sum_p f_{i_1 p} a_p^\dagger \right) \cdots \left(\sum_t f_{i_N t} a_t^\dagger \right) |0\rangle = \det(f_{ip})_{i,p \leq F},$$

the determinant of the *occupied block* of f : only terms with all $p \leq F$ survive the vacuum expectation value, and the sum over permutations reconstructs a determinant.

- The hypothesis $\langle c|\tilde{c}\rangle \neq 0$ says exactly that the occupied block is invertible, and intermediate normalisation rescales it to $\det(f_{ip}) = 1$. With f^{-1} (of the occupied block) available,

$$\sum_{i \leq F} f_{ki}^{-1} \tilde{b}_i^\dagger = \sum_i \sum_p f_{ki}^{-1} f_{ip} a_p^\dagger = a_k^\dagger + \sum_{p > F} \left(\sum_{i \leq F} f_{ki}^{-1} f_{ip} \right) a_p^\dagger,$$

the terms with $p \leq F$ having collapsed to δ_{kp} . Setting $C_{pk} = \sum_i f_{ki}^{-1} f_{ip}$ returns $b_k^\dagger$ exactly.

- A unitary (here: invertible) mixing of the $\tilde{b}_i^\dagger$ changes the determinant only by a factor, so $|\tilde{c}\rangle$ and $|c'\rangle$ are the same state. **The theorem is proved.**

What the hypothesis excludes

Determinants orthogonal to $|c\rangle$ – e.g. $|\Phi_i^\bullet\rangle$ itself – cannot be reached by $e^\tau |c\rangle$, whose overlap with $|c\rangle$ is always one. They are “infinitely far away” in the amplitudes: $C_{ai} \rightarrow \infty$. For a local variation this is no restriction at all.

Numerical check

hartreefock.py, class ThoulessRotation: six orbitals, three particles, random amplitudes C_{ai} of size 0.3, everything in the $\binom{6}{3} = 20$ -dimensional determinant space of week 38.

$\max [\hat{T}_i, \hat{T}_j] $	2.8×10^{-17}	scale s	$ e^{s\hat{T}} c\rangle - (1 + s\hat{T}) c\rangle $	ratio
$\max \hat{T}_i^2 $	0	0.4	3.68×10^{-2}	
$ e^{\hat{T}} c\rangle - \prod_i b_i^\dagger 0\rangle $	5.6×10^{-17}	0.2	9.20×10^{-3}	4.00
$\langle c e^{\hat{T}} c\rangle$	1.000000000000	0.1	2.30×10^{-3}	4.00
det of the occupied block of the orbital matrix	1.000000000000	0.05	5.75×10^{-4}	4.00
		0.025	1.44×10^{-4}	4.00

- The rotated state is a determinant to machine precision, and its overlap with the reference is exactly one – it is never orthogonal to the state we started from: exactly the freedom Hartree-Fock explores.
- Halving the amplitudes divides the discarded piece $e^{\hat{T}} - (1 + \hat{T})$ by four: the $2p$ - $2h$ admixtures are second order, as claimed. The notebook repeats this with the self-contained bit code of week 38.

Pause to think: counting, and what the exponential does

Pause to think #4

(about 3 min – discuss with your neighbour)

1. Eight orbitals, four particles. How many complex parameters does Thouless' theorem give the neighbourhood of a determinant? How many does a general normalised state in the 70-dimensional space have? Where did the rest go?
2. Write out $e^{\hat{T}}|c\rangle$ for $N = 2$ explicitly. Which determinants appear, with which amplitudes? What does this say about the coupled-cluster ansatz $e^{\hat{T}_1}|c\rangle$ of chapter 10, and about why CIS gives nothing in a Hartree-Fock basis?

Answer to pause #4

1. $(n - N) \times N = 16$ complex amplitudes C_{ai} , against 69 for a general normalised state in 70 dimensions. The other 53 directions leave the manifold of determinants: $2p-2h$, $3p-3h$, $4p-4h$ amplitudes that are *not* products of $1p-1h$ ones. Hartree-Fock optimises 16 parameters of a 69-parameter problem; correlation is the other 53.
2. $e^{\hat{T}}|c\rangle = (1 + \hat{T}_1)(1 + \hat{T}_2)|c\rangle = |c\rangle + \sum_a C_{a1}|\Phi_1^a\rangle + \sum_b C_{b2}|\Phi_2^b\rangle + \sum_{ab} C_{a1}C_{b2}a_a^\dagger a_1 a_b^\dagger a_2|c\rangle$. The doubles appear with amplitudes that are *products* of singles amplitudes, $C_{12}^{ab} = C_{a1}C_{b2} - C_{b1}C_{a2}$: no freedom of their own. So $e^{\hat{T}_1}|c\rangle$ is just another determinant (Thouless), coupled cluster with singles only can never beat Hartree-Fock, $\hat{T}_2$ is the first correlating operator, and in a Hartree-Fock basis ($f_{ai} = 0$) the singles amplitudes start at second order. CIS returns E^{HF} for the same reason: the singles span the tangent space of the determinant manifold at its stationary point.

What Thouless' theorem buys us

Any nearby determinant is

$$|c'\rangle = |c\rangle + |\delta c\rangle, \quad |c'\rangle \simeq \left(1 + \sum_{ai} \delta C_{ai} a_a^\dagger a_i\right) |c\rangle \quad \text{to first order,}$$

and the stationarity condition $\langle \delta c | \hat{H} | c \rangle = 0$ becomes

$$\sum_{ai} \delta C_{ai}^* \langle c | a_i^\dagger a_a \hat{H} | c \rangle = 0 \quad \text{for arbitrary } \delta C \quad \implies \quad \langle \Phi_i^a | \hat{H} | \Phi_0 \rangle = f_{ai} = 0,$$

Thursday's condition for all pairs at once, and Brillouin's theorem.

- Thouless' theorem confirms what Thursday assumed: to first order the $1p$ - $1h$ excitations *exhaust* the neighbourhood of a determinant, so the single-orbital variations were the general case.
- Three derivations, one equation. What the third route adds is the **exact** form of the neighbourhood – and with it the ability to go to second order, which is the subject of the next lecture.

Many-body connection

The Thouless rotation is also how mean fields are *parametrised* in practice: the Lipkin model below rotates every orbital by the same angle, $C_{ai} = \tan(\alpha/2) \delta_{pq}$; deformed nuclei, symmetry-broken Hartree-Fock for Hubbard, and the BCS state of chapter 7 are all Thouless rotations of a symmetric reference. The unitary form $e^{\hat{T} - \hat{T}^\dagger}$ is the orbital-rotation step of every modern Hartree-Fock code.

Friday, lecture 4: Stability of the Hartree-Fock solution

Stationary is not minimal (section 6.10)

The variational condition $f_{ai} = 0$ guarantees only that $\langle c | \hat{H} | c \rangle$ is *stationary*. A stationary point may be a minimum, a maximum or a saddle – and it matters:

- a Hartree-Fock solution on a saddle is a poor reference for perturbation theory or coupled cluster, because the corrections it generates are organised about the wrong state;
- there is a direction in the space of determinants along which the energy *decreases*, so a better single determinant exists – usually one that breaks a symmetry of $\hat{H}$ which the reference respected.

The question

Is

$$\frac{\langle c' | \hat{H} | c' \rangle}{\langle c' | c' \rangle} \geq \langle c | \hat{H} | c \rangle = E_0$$

for every nearby determinant $|c'\rangle = |c\rangle + |\delta c\rangle$?

Thouless' theorem tells us what “every nearby determinant” means, and lets us expand to second order:

$$|c'\rangle = e^{\sum \delta C_{ai} a_a^\dagger a_i} |c\rangle = \left\{ 1 + \sum_{ai} \delta C_{ai} a_a^\dagger a_i + \frac{1}{2!} \sum_{abij} \delta C_{ai} \delta C_{bj} a_a^\dagger a_i a_b^\dagger a_j + \dots \right\} |c\rangle,$$
$$\langle c' | c' \rangle = 1 + \sum_{ai} |\delta C_{ai}|^2 + \mathcal{O}(\delta C^3)$$

(intermediate normalisation $\langle c' | c \rangle = 1$; the cross terms between 1 and the linear piece vanish since $\langle c | a_a^\dagger a_i | c \rangle = 0$, and the linear piece with itself gives the sum of squares because the $1p-1h$ states are orthonormal).

The energy to second order

Expand the numerator and use $f_{ai} = 0$ to kill the terms linear in δC :

$$\begin{aligned}\langle c' | \hat{H} | c' \rangle &= \langle c | \hat{H} | c \rangle + \sum_{ab > F} \sum_{ij \leq F} \delta C_{ai}^* \delta C_{bj} \langle c | a_i^\dagger a_a \hat{H} a_b^\dagger a_j | c \rangle \\ &\quad + \frac{1}{2!} \sum \delta C_{ai} \delta C_{bj} \langle c | \hat{H} a_a^\dagger a_i a_b^\dagger a_j | c \rangle + \frac{1}{2!} \sum \delta C_{ai}^* \delta C_{bj}^* \langle c | a_j^\dagger a_b a_i^\dagger a_a \hat{H} | c \rangle + \dots\end{aligned}$$

Each expectation value is a Wick calculation of week 37. With

$$\hat{H} = \sum_{pq} \langle p | \hat{h}_0 | q \rangle a_p^\dagger a_q + \frac{1}{4} \sum \langle pq | \hat{v} | rs \rangle a_p^\dagger a_q^\dagger a_s a_r$$

- **First line** ($1p$ - $1h$ to $1p$ - $1h$, the matrix element of week 38's CI matrix): collecting the surviving contractions in a Hartree-Fock basis,

$$E_0 \sum_{ai} |\delta C_{ai}|^2 + \sum_{ai} |\delta C_{ai}|^2 (\varepsilon_a - \varepsilon_i) - \sum_{ijab} \langle aj | \hat{v} | bi \rangle \delta C_{ai}^* \delta C_{bj}.$$

The first term is E_0 times the norm correction; the second is the unperturbed excitation energy; the third is the particle-hole interaction.

- **Second line** (reference to $2p$ - $2h$): only $\hat{V}_N$ contributes,

$$\frac{1}{2} \sum_{ijab} \langle ab | \hat{v} | ij \rangle \delta C_{ai}^* \delta C_{bj}^* \quad (\text{and the third line is its Hermitian conjugate}).$$

The stability matrix

Define the two antisymmetrised blocks and the diagonal matrix of excitation energies,

$$A_{ai,bj} = -\langle aj|\hat{v}|bi\rangle_{\text{AS}}, \quad B_{ai,bj} = \langle ab|\hat{v}|ij\rangle_{\text{AS}}, \quad \Delta_{ai,bj} = (\varepsilon_a - \varepsilon_i) \delta_{ab} \delta_{ij}.$$

Then, dividing by the norm $1 + \sum |\delta C|^2$,

$$\frac{\langle c'|\hat{H}|c'\rangle}{\langle c'|c'\rangle} = E_0 + \frac{\Delta E}{1 + \sum_{ai} |\delta C_{ai}|^2}, \quad \Delta E = \frac{1}{2} \langle \chi | \hat{M} | \chi \rangle, \quad \chi = \begin{bmatrix} \delta C \\ \delta C^* \end{bmatrix}, \quad \hat{M} = \begin{pmatrix} \Delta + A & B \\ B^* & \Delta + A^* \end{pmatrix}.$$

- The denominator is positive, so **the Hartree-Fock solution is a local minimum if and only if $\Delta E = \frac{1}{2} \langle \chi | \hat{M} | \chi \rangle \geq 0$ for every χ** , i.e. if and only if every eigenvalue of $\hat{M}$ is non-negative.
- A necessary (not sufficient) condition, from χ restricted to the upper block: $\Delta + A \geq 0$ as a matrix. Cheap, and the usual first check.
- Why does χ contain both δC and δC^* ? Because the second-order energy has terms $\delta C^* \delta C$ (the A block), $\delta C \delta C$ and $\delta C^* \delta C^*$ (the B blocks): a quadratic form in the *real* variables $\text{Re } \delta C$, $\text{Im } \delta C$, written compactly in the complex ones. Thursday's one-parameter family is the 1×1 case, with $B_{ai,ai} = \langle aa|\hat{v}|ii\rangle = 0$ by antisymmetry.

The stability matrix is the RPA matrix

Many-body connection

$\Delta + A$ is exactly the Hamiltonian restricted to the $1p\text{-}1h$ space, measured from E_0 : the matrix of the Tamm-Dancoff approximation and of CIS. The full $\hat{M}$, with its B blocks, is the matrix of the random-phase approximation (chapter 7). So “the Hartree-Fock solution is stable” and “the RPA frequencies are real” are one and the same statement, and an imaginary RPA frequency is not a numerical accident but a signal that the mean field itself is wrong. TDA and RPA are the second derivative of the Hartree-Fock energy surface.

- In practice: solve the Hartree-Fock equations, then diagonalise $\Delta + A$ (cheap, necessary) and, if in doubt, $\hat{M}$ (sufficient). A negative eigenvalue names the *direction* χ in which to deform the reference – which is how symmetry-broken solutions are found.
- Two examples follow: one where $\hat{M}$ goes negative at a critical coupling (Lipkin), and one where it never does but the mean field is nevertheless useless (pairing).

An explicit instability: the Lipkin model (section 6.11)

N particles in two N -fold degenerate levels separated by ε , $W = 0$:

$$\hat{H} = \varepsilon \hat{J}_z + \frac{V}{2} (\hat{J}_+^2 + \hat{J}_-^2), \quad \hat{J}_z = \frac{1}{2} \sum_p (a_{p+}^\dagger a_{p+} - a_{p-}^\dagger a_{p-}), \quad \hat{J}_+ = \sum_p a_{p+}^\dagger a_{p-}.$$

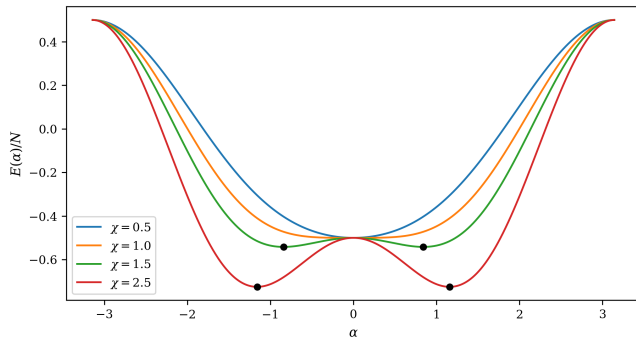
The natural mean field rotates every single-particle state by the same angle in the two-level space – a Thouless rotation with $C_{ai} = \tan(\alpha/2) \delta_{pq}$, normalised:

$$b_p^\dagger = \cos \frac{\alpha}{2} a_{p-}^\dagger + \sin \frac{\alpha}{2} a_{p+}^\dagger,$$

$$\frac{E(\alpha)}{N} = -\frac{\varepsilon}{2} \cos \alpha - \frac{\varepsilon \chi}{4} \sin^2 \alpha, \quad \chi = \frac{|V|(N-1)}{\varepsilon}$$

- $\frac{1}{N} \frac{dE}{d\alpha} = \frac{\varepsilon}{2} \sin \alpha (1 - \chi \cos \alpha)$: two stationary points, the *spherical* solution $\alpha = 0$ (always) and a *deformed* solution $\cos \alpha = 1/\chi$ (only for $\chi \geq 1$).
- $\frac{1}{N} \frac{d^2 E}{d\alpha^2} \big|_{\alpha=0} = \frac{\varepsilon}{2} (1 - \chi)$ changes sign at $\chi = 1$: the spherical solution turns from a minimum into a maximum exactly where the deformed one appears – a continuous symmetry-breaking transition.
- At $\alpha = 0$ the energy is $-\varepsilon/2$ per particle for *every* coupling: the interaction changes J_z by two units and has no expectation value in the undeformed determinant. **What changes with χ is the curvature, not the energy** – and no energy criterion would notice.
- The deformed minima come in pairs $\pm\alpha$ (the mean field has broken a symmetry of $\hat{H}$) and their energy is $-\frac{\varepsilon}{4}(\chi + 1/\chi)$ per particle: at $\chi = 1.5$ only 0.042ε below the spherical state.

The Lipkin energy surface



$E(\alpha)/N$ against the rotation angle, $N = 4$, $\varepsilon = 1$, $W = 0$.

- All curves pass through $-\varepsilon/2$ at $\alpha = 0$.
- $\chi = 0.5$: a simple well.
- $\chi = 1$: the curvature has vanished, the surface is quartic at the origin.
- $\chi = 1.5, 2.5$: $\alpha = 0$ is a local maximum, with two degenerate deformed minima at $\alpha = \pm 0.841$ and ± 1.159 , energies -0.542 and -0.725 .

Almost everything the stability analysis does algebraically can be read off this picture. The next slide does it algebraically anyway, because for a realistic system there is no α to plot against.

The stability matrix says the same, without the parametrisation

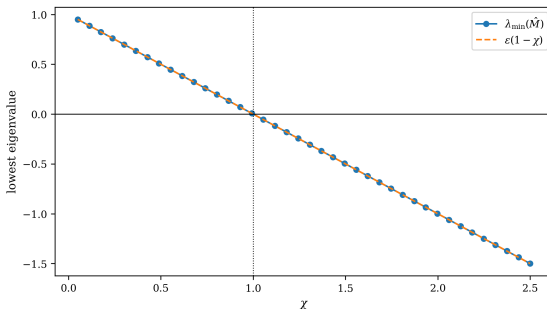
- The interaction changes J_z by two units, so it has no matrix element within the $1p-1h$ space: $A = 0$ and $\Delta + A = \varepsilon 1$.
- B has elements $\pm V$ connecting different pairs (a pair p raised and a pair q raised is the $2p-2h$ state J_+^2 makes); its largest eigenvalue is $|V|(N-1)$.
- Hence

$$\lambda_{\min}(\hat{M}) = \varepsilon - |V|(N-1) = \varepsilon(1 - \chi)$$

which is the second derivative of the previous slide up to the factor $N/2$.

hartreefock.py builds $\hat{M}$ numerically from $\hat{H}$ in the determinant basis ($N = 4$, $\varepsilon = 1$):

χ	$\lambda_{\min}(\hat{M})$	$\alpha_{\min}$	E_{mf}	E_{exact}	solution
0.25	0.75	0	-2.000000	-2.020726	spherical
0.50	0.50	0	-2.000000	-2.081666	spherical
0.90	0.10	0	-2.000000	-2.253886	spherical
1.00	0.00	0	-2.000000	-2.309401	critical
1.50	-0.50	0.841	-2.166667	-2.645751	deformed
2.00	-1.00	1.047	-2.500000	-3.055050	deformed



The numerical eigenvalues sit on the line $\varepsilon(1 - \chi)$ over the whole range and keep falling beyond the transition: a negative eigenvalue is a direction along which the energy decreases without limit until the nonlinearity of the parametrisation catches it.

Pause to think: stable, and useless

Pause to think #5

(about 4 min – discuss with your neighbour)

1. For $\chi = 2$ the spherical Lipkin determinant satisfies $f_{ai} = 0$ exactly and obeys Brillouin's theorem. Is it a Hartree-Fock solution? What goes wrong if one builds second-order perturbation theory on it?
2. The pairing model of weeks 37–38 (four doubly degenerate levels, four particles, $\varepsilon_p = p - 1$, pairing strength g): last week we saw that Hartree-Fock does nothing for it. Is the reference *stable*? Guess $\lambda_{\min}(\hat{M})$ as a function of g before looking at the next slide. Hint: which $1p-1h$ pairs does B connect, and what is Δ for them?

Answer to pause #5

1. It is a genuine stationary point of the energy on the manifold of determinants, so it satisfies the Hartree-Fock *equations* – but it is a saddle, not a minimum, and a local minimum is what the variational principle asked for. Second-order perturbation theory about it has the form $\sum |\langle \Phi_{ij}^{ab} | \hat{V} | \Phi_0 \rangle|^2 / (E_0 - E_{ij}^{ab})$; in the unstable channel the relevant “excitation energy” is governed by $\lambda_{\min} < 0$, the denominators are small and of the wrong sign, and the series is organised about the wrong state. The deformed solution is 0.5 lower at $\chi = 2$, but still 0.55 above the exact answer, and has broken a symmetry that must eventually be restored.

Answer to pause #5

2. $A = 0$: the pairing interaction cannot scatter a single particle-hole pair, so $\Delta + A$ is diagonal with $\Delta = \varepsilon_q - \varepsilon_p + g/2$ for the pair $p \rightarrow q$ (breaking the pair at p costs its pairing energy; $\lambda_{\min}(\Delta + A) = 1 + g/2$, so the necessary condition holds). B couples $\delta C_{q\uparrow, p\uparrow}$ to $\delta C_{q\downarrow, p\downarrow}^*$ through $\langle q\uparrow q\downarrow | \hat{v} | p\uparrow p\downarrow \rangle = -g/2$, so $\hat{M}$ falls apart into 2×2 blocks $\begin{pmatrix} \Delta & -g/2 \\ -g/2 & \Delta \end{pmatrix}$ with eigenvalues $\varepsilon_q - \varepsilon_p$ and $\varepsilon_q - \varepsilon_p + g$: moving the pair *coherently* costs exactly the level spacing, the interaction cancels out, and $\lambda_{\min}(\hat{M}) = \varepsilon_3 - \varepsilon_2 = 1$ for every g (numerically 1.0000 at $g = 0.25, \dots, 3$). The reference is stable, and Hartree-Fock – any number-conserving mean field – can never capture pair correlations. The mean field that does is the BCS state of chapter 7, a Thouless-like rotation that breaks particle-number symmetry instead.

Two lessons, and where this goes

Lesson 1: satisfying the Hartree-Fock equations is not enough

A solution can satisfy $f_{ai} = 0$ perfectly, obey Brillouin's theorem, and be a saddle. Only the stability matrix – equivalently, the RPA frequencies – tells the difference, and one needs the criterion, not a plausibility argument: at $\chi = 1.5$ the deformed state is a mere 0.042ϵ per particle below the spherical one.

Lesson 2: symmetry breaking is a device, and the bill comes later

The deformed solution buys a lower energy by breaking a symmetry the exact ground state does not break. It captures correlations cheaply – and the symmetry must eventually be restored, by projection or by a method that never broke it. Deformed nuclei, spin-density waves in the Hubbard model and BCS superconductivity are all this mechanism.

- Stable but useless (pairing model): the mean field is fine, the physics is in the doubles; perturbation theory and coupled cluster start there.
- Unstable (Lipkin, $\chi > 1$; Hubbard beyond a critical U): the mean field itself must change before any correlation method makes sense.
- In both cases the diagnosis is the same object, $\hat{M}$, and its eigenvalue problem is chapter 7.

Exercises: week 39, and answers for week 38

Exercise session: week 39 (suggested set)

Three exercises, one per derivation of the week. Full answers to the parts taken from the chapter exercise sections are in the lecture notes; the new parts are answered on next week's slides.

Exercise 1: varying one orbital (section 6.6; new)

- a) Let $\delta\phi_i = \sum_{\lambda} d_{\lambda} \phi_{\lambda}$ be arbitrary. Show, using linearity of the determinant in row i , that the components with $\lambda \leq F$ change Φ_0 by at most a factor, and that the normalisation constant is $N_i^{-2} = 1 + \sum_{a>F} |d_a|^2$ once the occupied components are dropped.
- b) Derive $E(\eta) = E_0 + 2 \operatorname{Re}(\eta f_{ia}) + |\eta|^2 (\langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle - E_0) + \mathcal{O}(|\eta|^3)$ from the exact expression with the denominator $1 + |\eta|^2$, and show that the cubic term is $-2|\eta|^2 \operatorname{Re}(\eta f_{ia})$. What does a real η alone fail to fix?
- c) With $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$ normal ordered with respect to $|\Phi_0\rangle$, show that $\langle \Phi_0 | \hat{V}_N | \Phi_i^a \rangle = 0$ by writing $\hat{V}_N$ in its particle-hole form (week 37) and letting each term act on $|\Phi_i^a\rangle$, without Wick's theorem. Then show that $f_{ai} = 0$ for all pairs makes f block diagonal, and that diagonalising the blocks does not change $|\Phi_0\rangle$.
- d) For the random $n = 8$, $N = 4$ Hamiltonian of the week-38 notebook, plot $E(\eta)$ for real $\eta \in [-0.15, 0.15]$ and the pair with the largest $|f_{ia}|$, in the original basis and in the Hartree-Fock basis. Check that the odd part of $E(\eta)$ equals $2\eta f_{ia}$ up to the cubic term, and that a purely imaginary η produces no linear change.

Exercise session: week 39 (continued)

Exercise 2: single-particle energies and Thouless' theorem (chapter 6, exercises 2b–c and 3a, 3b, 3d)

- a) Show that $E^{\text{HF}} \neq \sum_i \epsilon_i^{\text{HF}}$ and derive $E^{\text{HF}} = \sum_{i \leq F} \epsilon_i^{\text{HF}} - \frac{1}{2} \sum_{ij \leq F} \langle ij | \hat{v} | ij \rangle$. Verify it numerically for the trap system ($N = 4$) or the random Hamiltonian.
- b) Koopmans: compute the energy of the determinant with orbital k removed, with the remaining orbitals frozen, and show that it is $E^{\text{HF}} - \epsilon_k^{\text{HF}}$. Why is the corresponding statement for *adding* a particle in orbital a less accurate?
- c) Prove $[a_a^\dagger a_i, a_b^\dagger a_j] = 0$ from the anticommutation relations, marking where the disjointness of the particle and hole index sets is used. Show $\hat{T}_i^2 = 0$ and hence $e^{\hat{T}} = \prod_i (1 + \hat{T}_i)$; write out the $2p$ - $2h$ content of $\hat{T}_i \hat{T}_j$.
- d) Show $\langle c | \tilde{c} \rangle = \det(f_{ip})_{i,p \leq F}$ for $\tilde{b}_i^\dagger = \sum_p f_{ip} a_p^\dagger$, first for $N = 2$ explicitly, then in general, and explain why $\langle c | \tilde{c} \rangle \neq 0$ is exactly what the proof needs.

Exercise session: week 39 (continued)

Exercise 3: stability (chapter 6, exercise 5a–c, plus the pairing model)

- a) Derive the norm $\langle c'|c' \rangle = 1 + \sum_{ai} |\delta C_{ai}|^2 + \mathcal{O}(\delta C^3)$ of the Thouless-rotated determinant, keeping track of which terms are second order.
- b) From the second-order energy, obtain $\Delta E = \frac{1}{2} \langle \chi | \hat{M} | \chi \rangle$ and the blocks Δ , A , B . Why does χ contain both δC and δC^* ? Show that $\hat{M}$ is Hermitian and that restricting χ to its upper half gives the necessary condition $\Delta + A \geq 0$.
- c) Lipkin model, $W = 0$: show $A = 0$, show that the largest eigenvalue of B is $|V|(N - 1)$, and recover $\lambda_{\min}(\hat{M}) = \varepsilon(1 - \chi)$.
- d) Pairing model, four levels, four particles: show analytically that $\hat{M}$ falls apart into 2×2 blocks with eigenvalues $\varepsilon_q - \varepsilon_p$ and $\varepsilon_q - \varepsilon_p + g$, so that $\lambda_{\min}(\hat{M}) = 1$ for every g , and confirm it with the notebook. What does this say about the prospects of a number-conserving mean field for pairing?

Answers to the week-38 exercises: 1, bit patterns

1a) Five orbitals, bits $\alpha_0 \dots \alpha_4$, α_0 leftmost

- $a_3^\dagger a_1 |01011\rangle$: occupied $\{1, 3, 4\}$. a_1 : no occupied state below 1, phase $+$, gives $|00011\rangle$. Then $a_3^\dagger$ on a state in which 3 is *occupied*: **zero** (Pauli).
- $a_0^\dagger a_4 |01101\rangle$: occupied $\{1, 2, 4\}$. a_4 : two occupied states below 4, phase $(-1)^2 = +$, gives $|01100\rangle$; $a_0^\dagger$: none below, phase $+$; result $+|11100\rangle$. “Between” rule: occupied states strictly between 0 and 4 are $\{1, 2\}$, two of them, phase $+$. Same answer.
- $a_4^\dagger a_3^\dagger a_1 a_0 |11010\rangle$: occupied $\{0, 1, 3\}$. a_0 then a_1 (rightmost first) give $+|00010\rangle$; $a_3^\dagger$ on occupied 3: **zero**.

Two of the three vanish – the lesson is to check occupancy before counting signs. (Replace $a_3^\dagger$ by $a_2^\dagger$ in the first: occupied $\{1, 3, 4\} \rightarrow \{3, 4\} \rightarrow \{2, 3, 4\}$, no occupied state between 1 and 2, answer $+|00111\rangle$.)

Answers to the week-38 exercises: 1, continued

1b) The phase of $a_j^\dagger a_i$

a_i contributes $(-1)^{n_{<}(i)}$, $n_{<}(i)$ the number of occupied states below i ; $a_j^\dagger$ then contributes $(-1)^{n'_{<}(j)}$ counted in the state *after* the removal. For $i < j$: $n'_{<}(j) = n_{<}(j) - 1$ and $n_{<}(j) = n_{<}(i) + 1 + n_{(i,j)}$ (i itself and the $n_{(i,j)}$ occupied states strictly between), so the total exponent is $2n_{<}(i) + n_{(i,j)}$. For $j < i$: $n'_{<}(j) = n_{<}(j)$ and $n_{<}(i) = n_{<}(j) + n_{(j,i)}$ (j is empty), exponent $2n_{<}(j) + n_{(j,i)}$. Either way $(-1)^l$ with l the number of occupied states strictly between. The argument uses $i \neq j$ when it says that j is empty after the removal and that i lies on one side of j ; for $i = j$ the operator is the number operator with phase $+1$.

1c) Counting

$a_p^\dagger a_q^\dagger a_s a_r |\Phi\rangle \neq 0$ needs $r \neq s$ both occupied – $N(N-1)$ ordered pairs – and then $p \neq q$ both empty in the state with r, s removed: $(n - N + 2)(n - N + 1)$. Total $N(N-1)(n - N + 2)(n - N + 1)$: for $n = 40$, $N = 8$ that is $56 \times 1122 = 62\,832$ against $n^4 = 2\,560\,000$, i.e. 2.5%. `apply_hamiltonian` on random patterns with $n = 8$, $N = 4$ finds $360 = 12 \times 30$ non-zero quadruples, as the formula says.

Answers to the week-38 exercises: 2, the FCI equation from the variational principle

2a) Rayleigh quotient

With C and C^* independent, $\frac{\partial}{\partial C_K^*} \frac{C^\dagger H C}{C^\dagger C} = \frac{(H C)_K}{C^\dagger C} - \frac{C^\dagger H C}{(C^\dagger C)^2} C_K = \frac{(H C - E C)_K}{C^\dagger C}$, which vanishes for all K iff $H C = E C$: the stationary points are the eigenvectors and the stationary values the eigenvalues. With a multiplier, $\partial[C^\dagger H C - \lambda(C^\dagger C - 1)]/\partial C_K^* = 0$ gives $H C = \lambda C$, and contracting with $C^\dagger$, $\lambda = C^\dagger H C = E$.

2b) Two levels, one pair

$$H = \begin{pmatrix} -g/2 & -g/2 \\ -g/2 & 2\xi - g/2 \end{pmatrix}, \det(H - E I) = (E + g/2)(E - 2\xi + g/2) - g^2/4 = 0, \text{ so } E_{\pm} = \xi - \frac{g}{2} \pm \sqrt{\xi^2 + \frac{g^2}{4}}.$$

Expanding the root, $E_- = -\frac{g}{2} - \frac{g^2}{8\xi} + \frac{g^4}{128\xi^3} - \dots$. Second-order perturbation theory: $E_{\text{ref}} = -g/2$ and

$E^{(2)} = \frac{|\langle \Phi_1 | \hat{H} | \Phi_0 \rangle|^2}{E_{\text{ref}} - H_{11}} = \frac{g^2/4}{-2\xi} = -\frac{g^2}{8\xi}$: the expansion of the exact root reproduces it term by term, and the next term, $+g^4/128$, is fourth order (there is no third-order term for this 2×2 problem). At $\xi = g = 1$: $E_- = -0.618034$, second order -0.625 .

Answers to the week-38 exercises: 2, continued

2c) Characteristic polynomials

For the 6×6 seniority-zero matrix at $g = 1$ (basis $|12\rangle, |13\rangle, |14\rangle, |23\rangle, |24\rangle, |34\rangle$, diagonal 1, 3, 5, 5, 7, 9, off-diagonal $-\frac{1}{2}$ between pairs sharing a level),

$$p(E) = E^6 - 30E^5 + 352E^4 - 2038E^3 + 5979E^2 - 7940E + 3100,$$

trace = 30 and det = 3100 by hand, the rest with `numpy.poly`; its roots 0.635548, 2.935381, 5, 5, 7.208940, 9.220130 agree with `eigh` to 10^{-13} (the double root 5 is the pair $|14\rangle \pm |23\rangle$, which the interaction does not split). For the 70×70 matrix the coefficients span 55 orders of magnitude (largest 9.6×10^{54}), the roots computed from the *exact* coefficients are off by up to 13, and a relative noise of 10^{-14} on the coefficients makes them garbage – while the same noise on the matrix moves the eigenvalues by 10^{-13} . The eigenvalue problem is well conditioned; the polynomial is not. This is why nobody solves $\det(H - E1) = 0$ beyond 3×3 .

Answers to the week-38 exercises: 2, interlacing

2d) Interlacing (Hylleraas-Undheim-MacDonald)

Diagonalising the leading $k \times k$ blocks in the order $|12\rangle, |13\rangle, |14\rangle, |23\rangle, |24\rangle, |34\rangle$:

k	$E_1^{(k)}$	$E_2^{(k)}$	$E_3^{(k)}$	$E_4^{(k)}$	$E_5^{(k)}$	$E_6^{(k)}$
1	1					
2	0.881966	3.118034				
3	0.794697	3.052662	5.152641			
4	0.708712	3.000000	5.000000	5.291288		
5	0.648907	3.000000	5.000000	5.102196	7.248898	
6	0.635548	2.935381	5.000000	5.000000	7.208940	9.220130

Every column decreases monotonically as k grows and $E_m^{(k)} \geq E_m^{(k+1)} \geq \dots \geq E_m$: the m -th Ritz value in a subspace is an upper bound on the m -th exact eigenvalue, not only the lowest. (Reading down a column also shows $E_m^{(k)} \leq E_{m+1}^{(k+1)}$, the interlacing proper.)

Answers to the week-38 exercises: 3, truncations and size consistency

3a) CIS and CID for the pairing model

Restricting the 70×70 matrix to excitation levels $\leq 0, 1, 2, 3, 4$ gives dimensions 1, 17, 53, 69, 70. CIS returns exactly E_{ref} because $\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0$: a single excitation breaks a pair and changes the seniority, which the pairing Hamiltonian conserves (week 38, exercise 2c of chapter 5). The reference is decoupled from the whole singles block, remains an exact eigenvector of the truncated matrix, and is its lowest eigenvalue for the couplings considered. CID keeps the reference and the four $2p$ - $2h$ seniority-zero states $|13\rangle, |14\rangle, |23\rangle, |24\rangle$ (plus $2p$ - $2h$ broken-pair states that decouple); it misses the $4p$ - $4h$ state $|34\rangle$, which is why the CID column of the week-38 table lies above FCI.

3b) Two non-interacting copies

Two two-level systems, one pair each, levels $0, \xi, 0, \xi$, the interaction acting within a copy only:

g	$2E(A)$	$E(AB)_{\text{FCI}}$	$E(AB)_{\text{CID}}$	CID error
0.25	-0.26556444	-0.26556444	-0.26550480	5.96×10^{-5}
0.50	-0.56155281	-0.56155281	-0.56066017	8.93×10^{-4}
1.00	-1.23606798	-1.23606798	-1.22474487	1.13×10^{-2}

Full CI is additive to machine precision ($E(A) = \xi - g/2 - \sqrt{\xi^2 + g^2/4}$ from 2b); the CID error grows by an order of magnitude each time g is doubled.

3c–d) The missing configuration, and the exponential

With $\hat{A}_2^X$ the operator exciting the pair of copy X , the exact state is $(1 + c\hat{A}_2^A)(1 + c\hat{A}_2^B)|\Phi_0\rangle = (1 + c\hat{A}_2^A + c\hat{A}_2^B + c^2\hat{A}_2^A\hat{A}_2^B)|\Phi_0\rangle$. The last term is a $4p\text{-}4h$ excitation of the combined reference and is absent from CID; adding the quadruples restores the exact energy, with the quadruple coefficient equal to c^2 , the *product* of the double amplitudes. With $\hat{T}_2 = t(\hat{A}_2^A + \hat{A}_2^B)$ the two terms commute (disjoint orbitals) and $(\hat{A}_2^X)^2 = 0$, so $e^{\hat{T}_2} = (1 + t\hat{A}_2^A)(1 + t\hat{A}_2^B)$: the exponential generates the product state with one parameter per copy, the same count as CID, and the energy is additive by construction (the notebook finds the FCI energy at $t = 0.2361$). Coupled cluster with doubles in miniature – and, today, the same mechanism as $e^{\hat{T}}$ in Thouless' theorem: products of low excitations generated for free.

What to do with the notebook

WeeklySlides/week39.ipynb contains, in this order (self-contained numpy, mirroring hartreefock.py; the bit code and SlaterBasis of week 38 are repeated so that nothing has to be imported):

1. The self-consistent field loop for the trap system: the density matrix as a projector, the energy against the orbital drift, Brillouin's theorem at convergence, $E^{\text{HF}} = \sum \epsilon_i - \frac{1}{2} \sum \langle ij | \hat{v} | ij \rangle$.
2. Varying one orbital: $a_a^\dagger a_i$ on the bit pattern against the “replace orbital i by a ” rule, $\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = f_{ia}$ for all pairs, $E(\eta)$ along real and imaginary η in the oscillator and the Hartree-Fock basis, the figure of Thursday.
3. Thouless' theorem: $[\hat{T}_i, \hat{T}_j] = 0$, $\hat{T}_i^2 = 0$, $e^{\hat{T}}|c\rangle$ against $\prod b_i^\dagger|0\rangle$, the overlap, the discarded piece scaling as the square of the amplitudes.
4. Stability: the matrix $\hat{M}$ from the Hamiltonian in the determinant basis; the Lipkin model ($\lambda_{\min}$ against χ , the energy surface); the pairing model ($\lambda_{\min} = 1$ for every g).
5. Numerical parts of the week-38 answers: the three bit-pattern examples, the quadruple count, the 2×2 and 6×6 characteristic polynomials, the interlacing table, the two-copy table.

Play with it: start the SCF loop from random orthonormal orbitals and count iterations; scale the interaction of the trap system up until the loop stops converging and see what the stability matrix says; for the Lipkin model add $W \neq 0$ and find the new critical coupling.

Summary of week 39

Three derivations, one equation

- Coefficients with multipliers: $h^{\text{HF}}C = \epsilon C$, a matrix that depends on its own eigenvectors; iterate to convergence of the *orbitals*
- One orbital, $\phi_i \rightarrow \phi_i + \eta \phi_a$, is $|\Phi_0\rangle + \eta a_a^\dagger a_i |\Phi_0\rangle$; complex η and the normal-ordered $\hat{H}$ give $\langle a | \hat{f} | i \rangle = 0$ – Brillouin's theorem derived, the eigenvalue form a convention on top
- The determinant: Thouless, $|c'\rangle = e^{\hat{T}}|c\rangle$ with $(n - N)N$ parameters and nothing else; first order gives $f_{ai} = 0$ again

In code: a loop of `eigh` on $h_0 + \rho \cdot v$; the check $\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = f_{ia}$ in the determinant basis; $\hat{M}$ from the same matrix – all in the notebook.

Second order: stability

- $\Delta E = \frac{1}{2} \langle \chi | \hat{M} | \chi \rangle$, $\hat{M} = \begin{pmatrix} \Delta + A & B \\ B^* & \Delta + A^* \end{pmatrix}$; minimum iff $\hat{M} \geq 0$; $\Delta + A \geq 0$ necessary
- $\hat{M}$ is the RPA matrix: unstable mean field = imaginary RPA frequency
- Lipkin: $\lambda_{\min} = \epsilon(1 - \chi)$, symmetry breaking at $\chi = 1$, energy insensitive, curvature not
- Pairing: stable for every g and useless – number-conserving mean fields never see pair correlations

Next week, and a closing remark

Next week (week 40)

What remains of chapter 6: the Baker-Campbell-Hausdorff expansion and the Trotter-Suzuki splitting that Thouless' theorem let us skip (sections 6.8–6.9), Hartree-Fock drawn as diagrams – the Fock operator as a bubble insertion and self-consistency as an infinite sum of bubbles (section 6.13) – and the one system where the whole programme is analytic, the homogeneous electron gas (section 6.12). Then the stability matrix becomes an eigenvalue problem of its own: the Tamm-Dancoff and random-phase approximations of chapter 7. Reading: chapter 6, sections 6.8–6.9 and 6.12–6.13; chapter 7; Szabo and Ostlund chapter 3 (the electron gas) and sections of Ring and Schuck on RPA.

This week one condition did all the work. Written with Lagrange multipliers it is an eigenvalue problem; written for one orbital it is $\langle a | \hat{f} | i \rangle = 0$, four symbols that empty the $1p$ - $1h$ block of every matrix in the rest of the course; written for the determinant it is the first derivative of the energy on the manifold of determinants, whose second derivative is the stability matrix and, next week, the RPA. Hartree-Fock is the choice of reference; everything after it is the choice of what to do with the $2p$ - $2h$ block that no single determinant can reach.